

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY**CHEMICAL PROCESS PRINCIPLES**
MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A**I. Answer all questions in one word or one sentence. Each question carries one mark.****(9 x 1 = 9 Marks)**

1	List any two refineries in south India.		
2	State batch processing.		
3	State Boyle's law.		
4	Identify the value of universal gas constant in kcal/(kmol·K).		
5	State the law of conservation of mass.		
6	Draw the block diagram for the unit operation crystallisation.		
7	Define the unit operation drying.		
8	Define Stoichiometric proportion.		
9	State limiting reactant.		

PART B**II. Answer any eight questions from the following. Each question carries 3 marks****(8 x 3 = 24 Marks)**

1	Summarize the history of chemical engineering as a discipline.		
2	List the duties of a chemical engineering technician.		
3	State Ideal gas equation and enumerate the terms involved in it.		
4	Gas contained in a closed vessel of volume 1 m ³ at a temperature of 298 K and pressure of 131.7 kPa is to be heated in such a way that, the pressure should not exceed 303.98 kPa. Compute the temperature of gas attained.		

5	Summarize the average molecular weight of a gas mixture.		
6	Illustrate the unit operation mixing.		
7	Sketch the block diagram for the unit operation evaporation.		
8	100 kg/h of a weak liquor containing 15% caustic (NaOH) by weight is to be concentrated in a pilot plant evaporator to get a thick liquor containing 40% by weight caustic (NaOH). Compute the amount of water to be evaporated in kg/hr.		
9	Summarize the recycle operation in a process plant with a block flow diagram.		
10	State excess reactant with an example.		

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	A solution of caustic soda contains 20% NaOH by weight. The density of the solution is measured to be 1.196 kg/l. Estimate the normality, molarity and molality of the solution. OR		
IV	Sodium chloride solution is prepared by dissolving 25 kg of sodium chloride in 100 kg of water. Compute the composition of the solution in (a) weight% and (b) mole%		
V	Estimate the density of air at STP, if the air contains 79% N ₂ and 21% O ₂ by volume. OR		
VI	A mixture of CH ₄ and C ₂ H ₆ has density 1.0 kg/m ³ at 273 K and 101.325 kPa pressure. Compute the mole% and weight% of CH ₄ and C ₂ H ₆ in the mixture.		
VII	In a laboratory experiment, 26.6 litres of NO ₂ at 80 kPa and 298 K is allowed to stand until the equilibrium is reached. At equilibrium, the pressure is measured to be 66.662 kPa. Estimate the partial pressure of N ₂ O ₄ in the final mixture. Given $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$ OR		
VIII	An inert gas of molecular weight 28 is admitted at a rate of 1 m ³ /min and at 202.65 kPa and 303 K to a pipe line in which natural		

	gas is flowing. The analysis of this gas at a very long distance shows that it contains 2.9% inert gas by volume. Estimate the flow rate of natural gas through the pipe line per min at 101.325 kPa and 303 K		
IX	The analysis of coal using in a boiler contains 63% carbon and 24% ash on weight basis. The analysis of refuse after combustion shows 7% carbon and rest ash. Estimate the percentage of the original carbon un-burnt in the refuse.		
	OR		
X	Feed to a continuous fractionating column analyses 28% benzene and 72% toluene by weight. The analysis of the distillate shows 52% benzene and 5% benzene by weight was found in the bottom product. Compute the amount of distillate and bottom product per 1000 kg of feed per hour.		
XI	A chemical industry requires acid containing 65% H_2SO_4 as raw material for processing and after processing the industry will get 25% H_2SO_4 as by product. The requirement for the industry is to have 1000 kg of 65% H_2SO_4 for daily operation. Estimate the amount of commercial grade sulphuric acid containing 98% H_2SO_4 is to be purchased on a daily basis to obtain the desired acid containing 65% H_2SO_4 , by mixing with the byproduct acid available.		
	OR		
XII	A multiple-effect evaporator system is having a design capacity to process 1000 kg per day of solid caustic soda when it concentrates weak liquor from 4 to 25% by weight caustic soda. When the same plant is fed with a 10% weak liquor and if it is concentrated to 50% (both on weight basis), estimate the capacity of the plant in terms of solid caustic soda. Assume that the water evaporating capacity to be the same in both cases.		
XIII	A furnace using butane as fuel is fed with 50 kmol/h of butane and 2000 kmol/h of air. Compute the percentage excess air used and composition of the gases leaving the furnace assuming complete combustion of butane.		
	OR		
XIV	Formaldehyde is manufactured from methanol in a catalytic reactor. The production rate of formaldehyde is 1000 kg/h. If the conversion of methanol is 65%, estimate the required feed rate of methanol.		